

Evaluating Explainable AI

From Quality Properties to Metrics

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A provocation

“SHAP is better than LIME.”

Better according to what?

Faithfulness? Stability? Compactness? Human usefulness?

Predictive performance is not explanation quality

Does f predict correctly? $\neq$ Does E describe f correctly?

- Predictive accuracy compares model outputs with target labels.
- Faithfulness concerns the explanation relative to model behavior.

What is being evaluated?

$$x \xrightarrow{f} f(x) \xrightarrow{E} e \qquad Q(e, f, x, D, \theta_Q)$$

- x : input; f : predictive model; E : explanation method; e : explanation.
- Q is a procedure with data, parameters, and assumptions.

We evaluate an explanation under a protocol, not in isolation.

A good explanation is not one thing

- Faithful but difficult to understand.
- Plausible but unfaithful to the model.
- Compact but incomplete.
- Stable but irrelevant to the user's task.

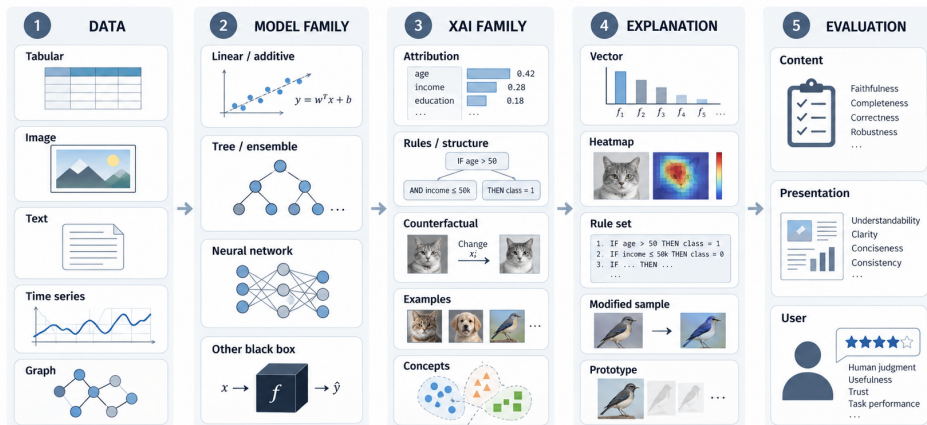
The scientific claim determines which trade-offs matter.

Data modality changes the evaluation operation

- Tabular: replace a feature or feature group.
- Image: mask a region; text: remove or replace a token.
- Time series: perturb an interval; graph: remove a node or edge.

The same property can require different operational metrics for different representations.

A schematic XAI landscape



Three evaluation perspectives

Functionally grounded Computational proxies; scalable and repeatable.

Human-grounded Simplified tasks with people; measures understanding or use.

Application-grounded Domain experts performing a real task; costly but direct.

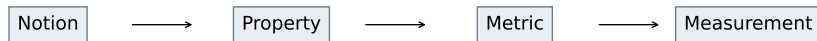
These answer different questions; one does not substitute for another.

The literature uses overlapping language

Explainability, interpretability, transparency, faithfulness, stability, robustness, comprehensibility, and actionability are related but not interchangeable.

A shared vocabulary makes claims testable.

Four levels



Notion Broad conceptual idea.

Property Desired quality of an explanation.

Metric Formal operationalization.

Measurement Observed value under a protocol.

From property to measurement

Claim

The explanation is faithful to the predictive model.

One operationalization

Deletion AUC; replace top-ranked features by the training median.

Measurement

0.31 on a named model, dataset, target, baseline, and seed.

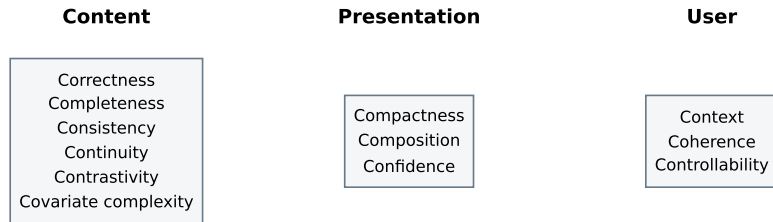
A scalar without its protocol is incomplete evidence.

The evaluation chain

Goal $\rightarrow$ Claim $\rightarrow$ Property $\rightarrow$ Metric $\rightarrow$ Protocol $\rightarrow$ Evidence

The metric is selected after the claim and property, not before them.

Co-12: a quality map, not a scorecard



Based on
the Content, Presentation, and User organization in the evaluation literature.[3]

Co-12: questions to ask

Content	Presentation / User
Correctness	Compactness: concise enough?
Completeness	Composition: appropriate form?
Consistency, continuity	Confidence: uncertainty represented?
Contrastivity	Context: useful for this task?
Covariate complexity	Coherence: aligns with domain knowledge?
	Controllability: user can interact?

One property, multiple metrics

Correctness / faithfulness

Deletion	Faithfulness correlation	Infidelity
Insertion	Randomization checks	Synthetic / white-box checks

Different operationalizations expose different assumptions and failure modes.[2]

Properties can conflict

A faithful explanation may be verbose.

A compact explanation may omit relevant behavior.

The evaluation objective must state which trade-offs are acceptable.

Faithfulness: perturb important features

Let $a = E(f, x)$, where a_i is feature importance. A perturbation-based test asks whether changing features with large $|a_i|$ changes $f(x)$.

$$\Delta_i = f(x) - f(x_{\setminus i})$$

The replacement rule is part of the metric.

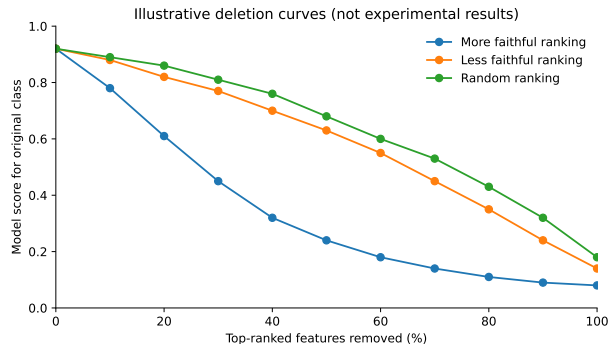
Faithfulness correlation

$$\text{FC}(x) = \text{corr}_S \left(\sum_{i \in S} a_i, f(x) - f(x_{\setminus S}) \right)$$

Positive association is evidence that attribution mass tracks output change under the sampled subsets and replacement protocol.

Deletion curve

Rank features by $|a_i|$, then progressively replace the most important ones. A steep score decrease early in the curve is consistent with a useful ranking under this deletion rule.



Deletion AUC and AOPC

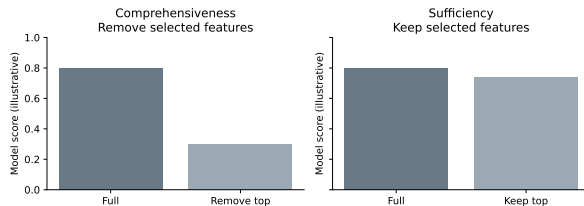
$$\text{AUC}_{del} = \int_0^1 f(x^{(\alpha)}) d\alpha \quad \text{AOPC} = \frac{1}{K} \sum_{k=1}^K [f(x) - f(x^{(k)})]$$

For a score-preserving output and a fixed deletion rule, lower deletion AUC or higher AOPC is generally preferred. Always state direction and target output.

Comprehensiveness and sufficiency

$$\text{Comp}(x, S) = f(x) - f(x_{\setminus S}) \quad \text{Suff}(x, S) = |f(x) - f(x_S)|$$

- Remove selected features: does the score change? Higher drop is generally preferred.
- Keep selected features: does the score remain? Smaller difference is generally preferred.

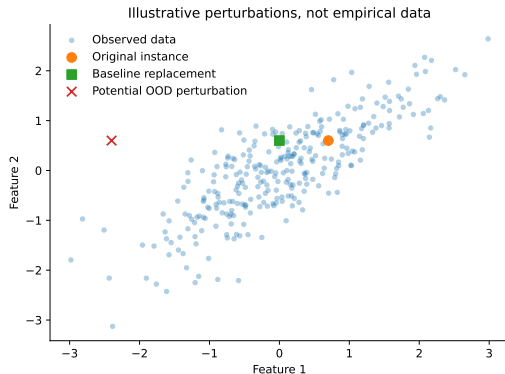


Infidelity

$$\text{Infid}(E, f, x) = \mathbb{E}_I \left[(I^T E(f, x) - (f(x) - f(x - I)))^2 \right]$$

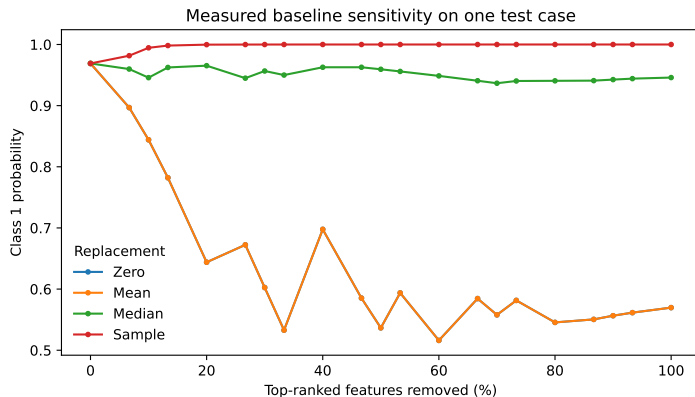
It compares attribution-predicted change with actual model change. The perturbation distribution I defines what changes are tested.[4]

The perturbation problem



- What does “feature absence” mean: zero, mean, median, or conditional sample?
- If replacement leaves the data manifold, are we measuring the explanation or model extrapolation?

Baseline sensitivity is a result



Measured on one held-out instance; the curve changes when the replacement convention changes.

Consistency is not continuity

Consistency Equivalent input/model conditions yield equivalent explanations.

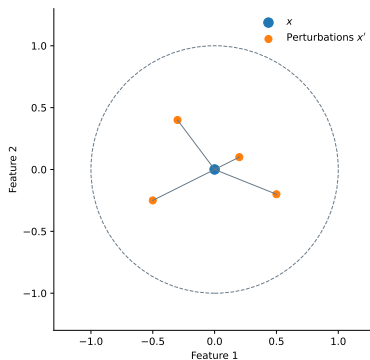
Continuity Similar inputs with similar predictions yield similar explanations.

Repeated seeds and equivalent implementations probe consistency; nearby inputs probe continuity.

Max-sensitivity

$$\text{Sens}_{max} = \max_{\|x' - x\| \leq r} \|E(f, x) - E(f, x')\|$$

The radius, input norm, explanation distance, and validity of perturbed inputs all matter.

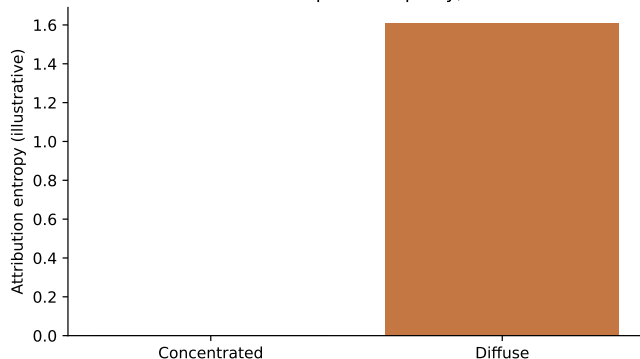


Compactness is not correctness

$$a = [0.91, 0.04, 0.03, 0.01, 0.01] \quad b = [0.21, 0.20, 0.20, 0.20, 0.19]$$

Which is easier to inspect? Does that make it more faithful?

Concentration is a compactness proxy, not correctness



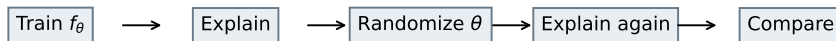
Entropy and sparsity

Normalize $p_i = |a_i| / \sum_j |a_j|$ and compute

$$H(a) = - \sum_i p_i \log p_i.$$

Lower entropy indicates concentration; higher Gini indicates concentration. Neither establishes that the explanation is correct or useful.

Sanity checks



If explanations barely change after model parameter randomization, model dependence is questionable. Passing a sanity check is necessary evidence, not proof of correctness.[1]

Faithful does not imply understandable

Faithful $\nRightarrow$ Understandable

A technically faithful explanation with a large number of predicates may be unusable for its audience.

Plausible does not imply faithful

Coherent with human knowledge $\nRightarrow$ Correct relative to model

A domain expert may find an explanation plausible even if the model relies on a spurious feature.

Human-grounded tasks

- Predict the model output; detect a model failure.
- Identify influential variables; answer comprehension questions.
- Measure decision time, confidence, and calibration.

Preference alone is weak evidence: “Which explanation do you like?” does not test task performance.



Application-grounded questions

Physician Does the explanation support diagnosis or error detection?

Engineer Does it help debug model behavior?

Auditor Does it reveal unacceptable dependencies?

Scientist Does it generate useful hypotheses?

A reproducible teaching benchmark

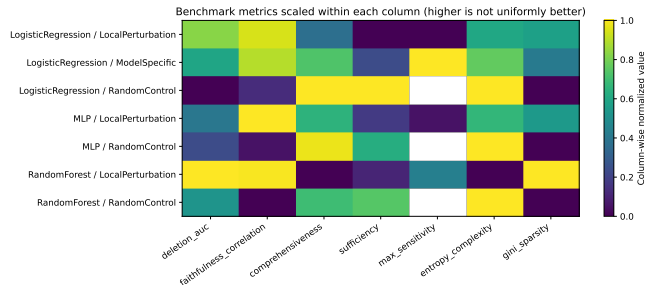
- Breast Cancer Wisconsin data; training-only standardization.
- Logistic regression, random forest, and MLP; class-1 probability.
- Local replacement attribution; model-specific attribution where available; random control.
- Median baseline; 25 selected test cases; explicit perturbation radius.

Designed for a short class run, not state-of-the-art comparison.

Metric matrix: keep directions visible

Deletion AUC ↓ Faithfulness correlation ↑
 Comprehensiveness ↑ Sufficiency ↓
 Max-sensitivity ↓ Gini sparsity: descriptive only

The table and plots below are computed from per-instance benchmark outputs. Column scaling is for visual comparison, not a common utility score.

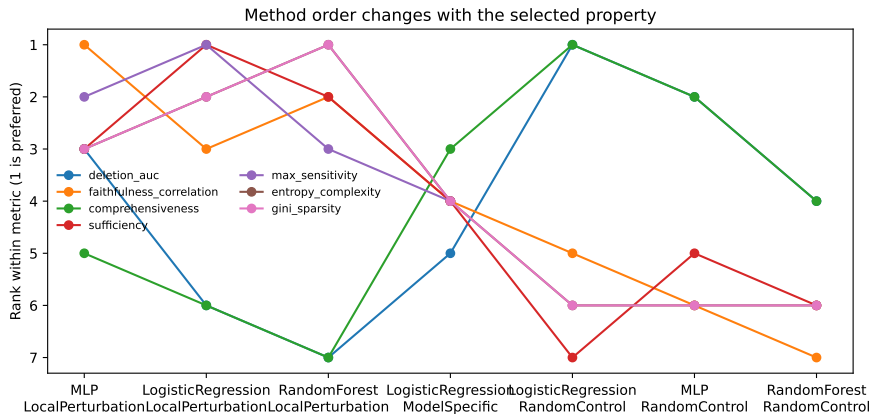


Include negative controls

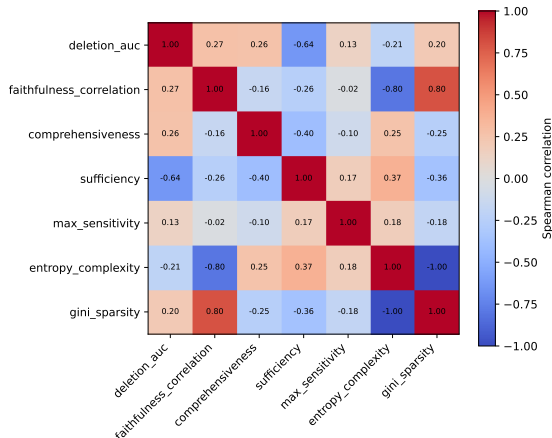
Random feature rankings and model randomization ask whether the protocol can reject explanations that should not earn high faithfulness.

An evaluation without controls can produce precise-looking but uncalibrated scores.

Metric disagreement

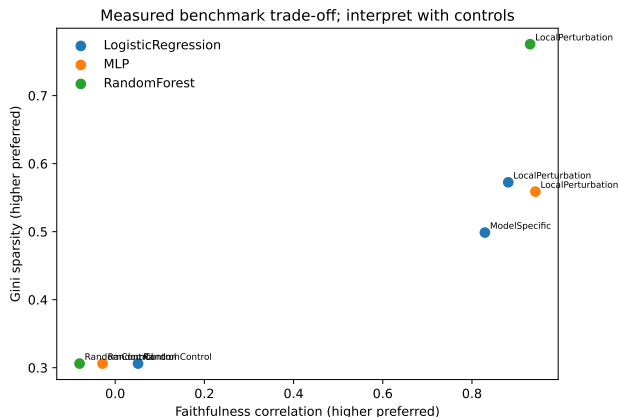


Metrics do not necessarily agree



Ranks and correlations describe this benchmark's models, samples, and protocol. They do not define a universal ordering.

Multi-objective view



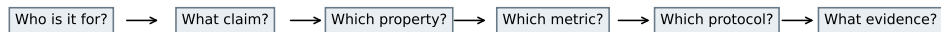
Different methods may trade off faithfulness, stability, and compactness. Avoid aggregation until the application defines a utility.

Discussion

- Better deletion AUC but worse stability: which method is better?
- If deletion is out of distribution, what did the score test?
- What control should a benchmark be able to reject?

Answer by naming the claim and intended use.

A defensible evaluation workflow



Then specify baselines, controls, aggregation, uncertainty, and the need for human evaluation.

Take-home points

1. Explanation quality is multidimensional.
2. A property and a metric are not the same thing.
3. Metrics encode assumptions about perturbations, baselines, and outputs.
4. Technical validity and human usefulness are different questions.
5. Strong evaluation combines complementary evidence and controls.

Final question

Do not evaluate an explanation.

Evaluate claims about an explanation.

Which claim? Which property? Which metric? Which protocol? Compared with what control?

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